THE PRUNING PARADOX

Why TRUNC() is the silent killer of enterprise database performance.

THE CORE CONFLICT

Understanding the wall between functions and the Oracle Optimizer.

The Blind Optimizer

Function Application

Applying `TRUNC()` transforms the key values. Oracle cannot logically map a transformed value back to the raw range-based metadata stored in the data dictionary.

Metadata Mismatch

Oracle stores partition "High Values" as raw data types. When functions wrap these keys, the optimizer loses its ability to prune, defaulting to a massive scan.

The Invisible Wall

Database engines are designed to optimize "Naked" predicates. Wrapping a key in a function creates a logical barrier that partition pruning cannot penetrate.

In a table with 1,000 partitions, a single `TRUNC()` call increases the scanning workload by 1,000x.

$$\text{ScanCost} = \sum_{i=1}^{N} P_i$$

Where N = Total Partitions instead of 1.

Syntactic Comparison

Metric	Non-Sargable (Bad)	Sargable (Good)
Syntax	`WHERE TRUNC(sale_dt) = :v`	`WHERE sale_dt >= :v AND sale_dt < :v + 1`
Plan Op	TABLE ACCESS FULL	PARTITION RANGE SINGLE
PSTART/PSTOP	1 / ALL	N / N
Optimizer	Heuristic / Blind	Cost-Based / Informed

Catastrophic Overhead

100x

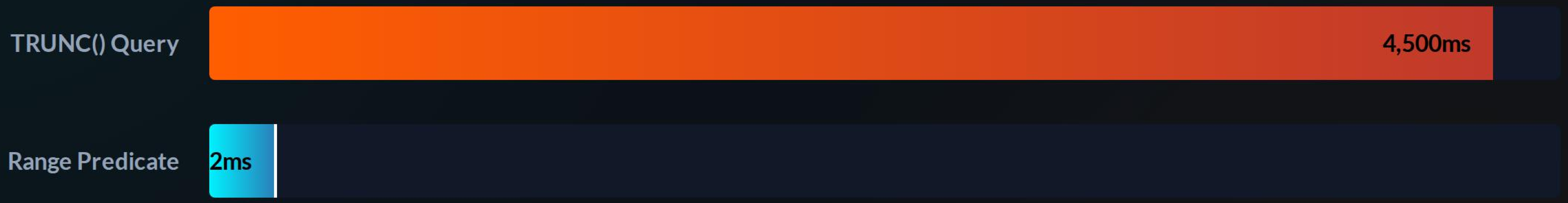
Excessive I/O Load

Efficiency Loss

On a 1TB partitioned table, failing to prune results in reading 1,000GB instead of 10GB. This saturates the I/O subsystem, pollutes the Buffer Cache, and triggers massive Latch Contention.

The business impact is high latency for critical reporting and transaction processing.

Performance Benchmark



Measured execution time on a table with 500 million rows across 365 partitions.

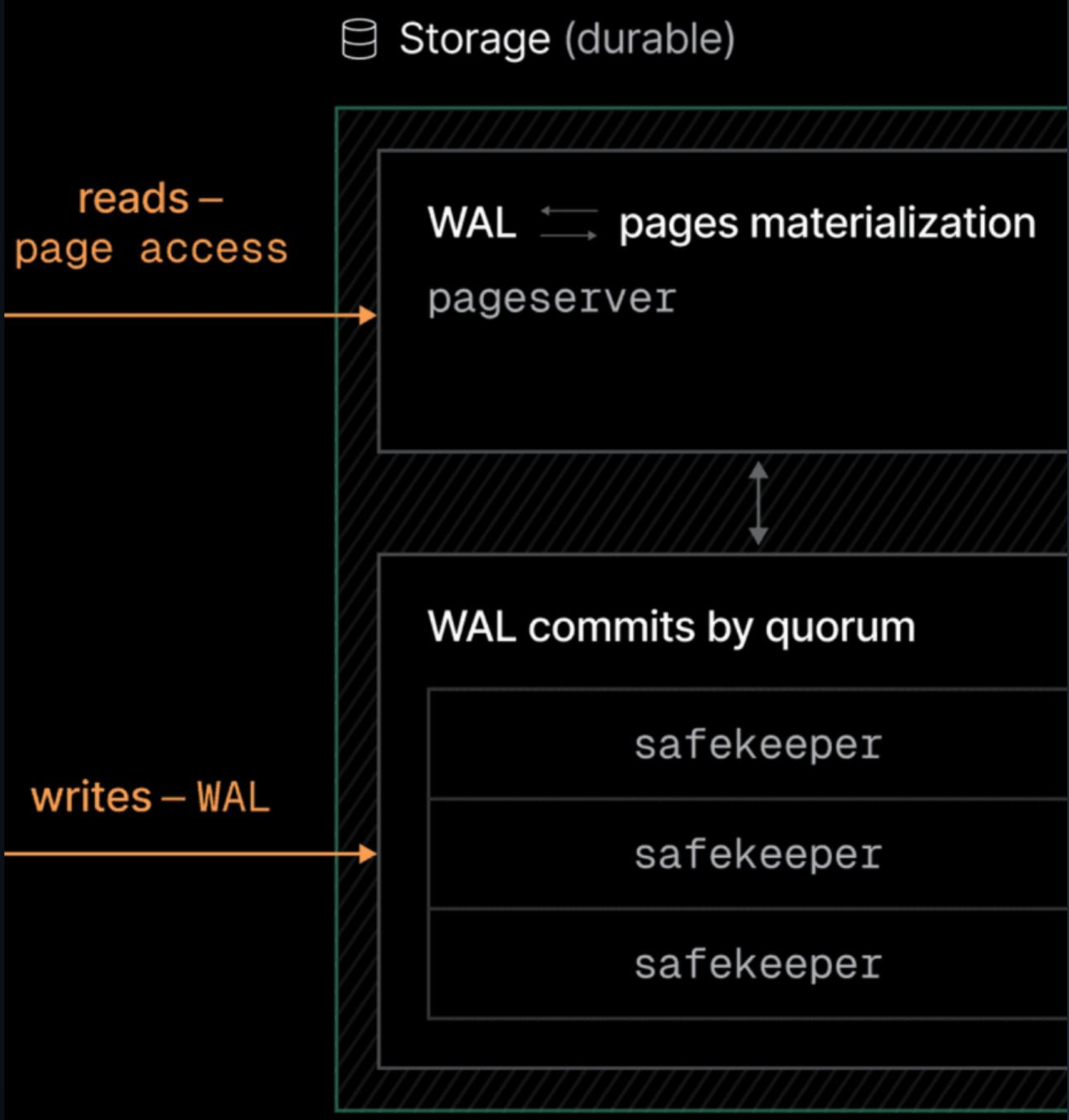
EXECUTION INTELLIGENCE

Decoding the truth inside DBMS_XPLAN.

The Smoking Gun

Don't guess—verify. When inspecting an Oracle execution plan, the **PSTART** and **PSTOP** columns are your primary evidence.

If you see "1" and "ALL", your partitioning strategy has failed. If you see specific numbers (e.g., 25 and 25), you have achieved **Partition Pruning**.



The Developer's Rulebook

- **Forbidden:** Do not use functions on partition keys in the WHERE clause. TRUNC, TO_CHAR, and UPPER will destroy pruning.
- **Sargable Fix:** Always rewrite predicates to use raw column ranges. Let the database compare values directly.
- **Inspection:** Use `EXPLAIN PLAN` or `DBMS_XPLAN.DISPLAY_CURSOR` to verify the PSTART/PSTOP markers.
- **Input Sanitation:** Cast your bind variables to the column's data type before the query hits the engine.

The Lifecycle of a Query

Syntax Parse

SQL is checked for basic grammatical correctness.

Predicate Check

CBO identifies sargable vs non-sargable conditions.

Pruning Decision

Engine identifies physical partitions for scan.

I/O Execution

Data blocks are retrieved from storage segments.

OPTIMIZING THE FUTURE

Thank you. Questions on Sargability, FBIs, or Oracle Optimization?

DBA_RESOURCES | PERF_TUNING_MASTERCLASS

Image Sources

https://static.vecteezy.com/system/resources/thumbnails/077/948/027/large/abstract-blue-digital-particle-wall-with-glowing-lines-and-floating-light-fragments-moving-in-depth-futuristic-technology-background-with-dynamic-data-stream-effect-video.jpg

Source: www.vecteezy.com

https://neon.com/_next/image?url=%2Fdocs%2Fintroduction%2Fneon-architecture-overview.png&w=1920&q=75&dpl=dpl_tJYWTHV13UEdhjqWJPK7MzSRa1rT

Source: neon.com